

ASTRONAUTS



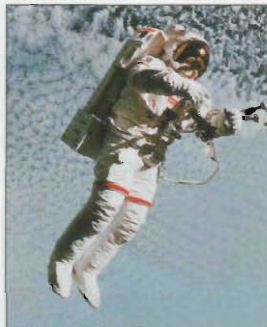
MORE THAN 350 PEOPLE have travelled into space; 26 on missions to the Moon and the rest in orbit around Earth. For journeying into space, astronauts must be physically and mentally fit. They must also be trained to prepare them for living and working in the hostile environment of space.

Spacesuit

When astronauts work outside the spacecraft, they need to wear a suit that keeps their body at the correct temperature and protects them from fast-moving micrometeoroids. The suit must also provide oxygen for breathing and be pressurized because there is no air or atmospheric pressure in space.

Yuri Gagarin

The first person to fly into space was a Russian, Yuri Gagarin (1934–68). His flight on 12 April 1961 took him once around the Earth and lasted 108 minutes. No one knew how the space flight would affect a human, so Gagarin's spacecraft, *Vostok 1*, was controlled from the ground.



MMU

To fly free from the spacecraft, an astronaut wears a powered backpack, the Manned Manoeuvring Unit (MMU). Mini nitrogen thrusters, operated from arm rests, propel the astronaut at about 20 m/s (65 ft/s).

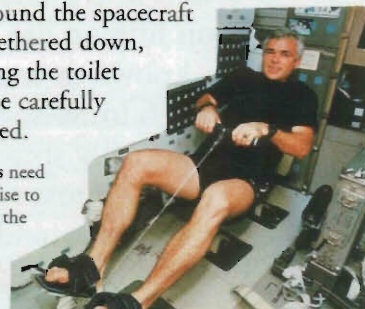


Apollo 9 spacesuit

Living in space

Daily life for an astronaut includes all the usual things, such as breathing, eating, sleeping, and going to the bathroom. The big difference, however, is living in weightless conditions. Sleeping astronauts float around the spacecraft unless tethered down, and using the toilet has to be carefully controlled.

Astronauts need daily exercise to keep fit in the weightless conditions of space.



Meal tray strapped to leg.

Vacuum-wrapped food pack

Rubber grips stop items floating away.

Space food

Meals on the space shuttle are prepared from 70 different foods and 20 drinks. The meal tray is strapped down and the food eaten with the hand or cutlery. Liquids are sucked from cartons or tubes.



Space toilet

Astronauts outside the spacecraft "go to the toilet" in their spacesuit, where the waste materials are collected. Inside the craft, they use a space toilet, making sure they are firmly strapped to the seat. The waste is sucked away by the toilet and collected in a secure unit.



Rubber suction cups



Suction shoes

Staying in one place in a spacecraft can be a problem. Suction-cup shoes allow astronauts to get a better grip.

Working in space

Each member of a space crew has specific tasks. These may include flying the craft, releasing a satellite into orbit, or testing new equipment. The weightless conditions of space mean that astronauts can also perform experiments not possible on Earth.

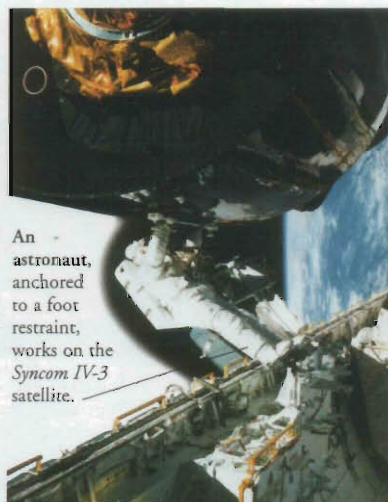


Repair work

Once a satellite is in space it is left to work on its own. But occasionally one needs repairing. The cargo bay of the space shuttle is equipped with a robotic arm, which specially trained astronauts use to recover the satellite. They can then repair the satellite and release it back into orbit.

Experiments

Astronauts have carried out many experiments in space. These include observing how living things such as bees are affected by weightlessness.



An astronaut, anchored to a foot restraint, works on the Syncom IV-3 satellite.

Endurance record

Most astronauts spend only a few days in space, but some stay for months. Russian cosmonaut Sergei Avdeyev holds the overall endurance record (748 days). Russian Valeri Poliakov holds the record for longest single stay (438 days).



Valeri Poliakov



Space animals

Humans are not the only space travellers; early ones included dogs, rats, and mice. Animals are no longer sent into space alone, but flies, frogs, and tadpoles occasionally accompany human astronauts.

Chimpanzee Ham returned safely from his 1961 flight.

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